

WASP-36b

R-0990

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_180304 · created 2026-07-29T23:18:30+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458181.8188 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.103 +/- 0.047
Transit depth (Rp/Rs)^2	1.06 +/- 0.96 [%]
Orbital Inclination (inc)	78.59 +/- 2.87
Airmass coefficient 1 (a1)	0.99 +/- 0.64
Airmass coefficient 2 (a2)	0.03 +/- 0.16
Scatter in the residuals of the lightcurve fit is	68.12 %
Best Comparison Star	None
Optimal Aperture	2.88
Optimal Annulus	3.75
Transit Duration (day)	0.021 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.103 ± 0.047 vs 0.1368 (deviation 0.62σ)
Duration fit vs published	0.021 d vs 0.0758 d (deviation 1.7σ)
Mid-time O–C	-4.7 min
Depth SNR	1.9
Residual scatter	68.12 %

Light curve

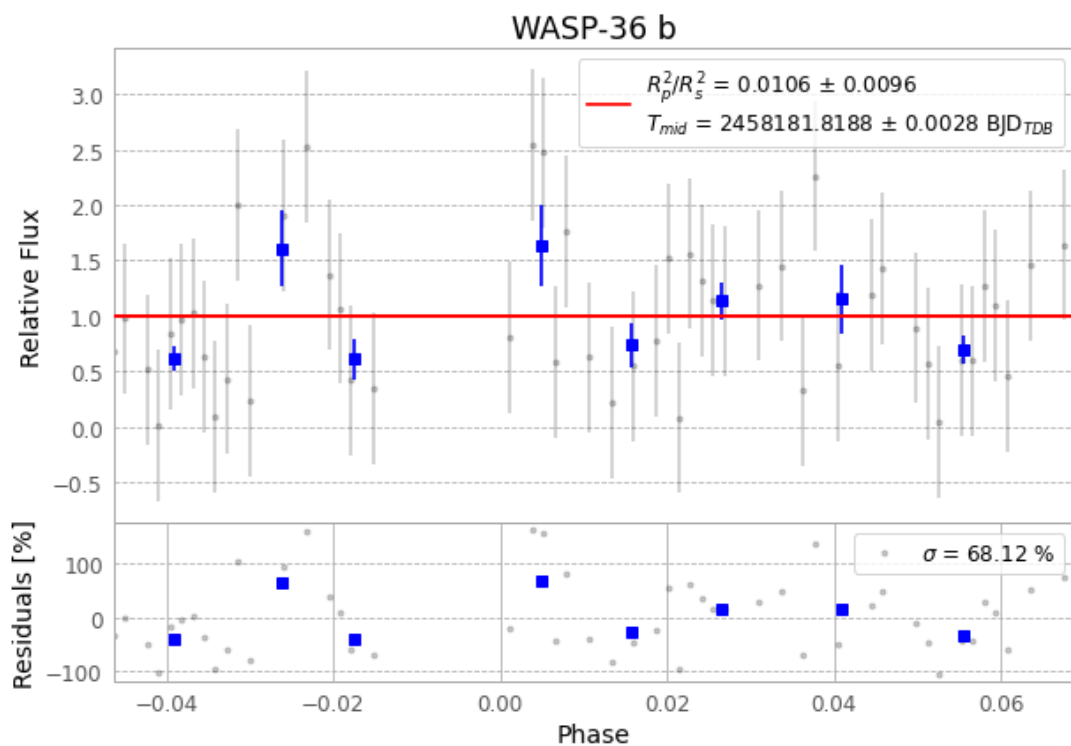


Figure 1 — FinalLightCurve_WASP-36 b_2018-03-03.png (SHA-256 687714f0e24ff8ca...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [441, 455], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 523d5322731d1dee...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

86 FITS frames (56 MB), WASP-36180304054812.FITS → WASP-36180304100312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-180304.log (55cc515dfb1f...), FinalLightCurve_WASP-36 b_2018-03-03.png (687714f0e24f...), FinalLightCurve_WASP-36 b_2018-03-03.csv (21be92a6bb62...)

Compute resources

Wall time 170.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 23:18Z.